Query generation

Write queries a user of the corpus could ask

task instruction t (optional)

3 documents d_1, d_2, d_3 sampled from the corpus

↓

LLM writer

↓

What non-genetic factors can trigger cancer development in the presence of oncogenic mutations?

$$q_j = \text{Write}(t, d_1, d_2, d_3)$$

kept if 3–4,000 characters; 100 per task

Retrieval pool

Every model retrieves its top 10 for every query

top 10 for q_j, by rank →

Qwen3-Embedd...
Linq-Embed-Mi...
all-MiniLM-L6-v2
baseline-bm25s
⋮

4 of 28 models; one colour per document retrieved by several

$$D_j = \bigcup_m \text{top10}_m(q_j)$$

here $|D_j| = 87$ documents, each judged once

Judging

An LLM judge rates what the models retrieve

Pointwise (default): grade each document 0 to 5

(q_j, d) → **judge** → 0 1 2 3 4 5 p(grade)

$$g(q_j, d) = \sum_k k \cdot p(k \mid q_j, d) = 4.3$$

Pairwise: which of two top 10s is better

$(q_j, \text{top10}_A, \text{top10}_B)$ → **judge** both orders, averaged

$$s_A = P(A \text{ wins}) + \tfrac{1}{2}P(\text{tie}) = 0.80$$

Scoring

Turn the judgments into one score per model

Pointwise

$$\text{nDCG@10} = \frac{\text{DCG of the model's top 10}}{\text{DCG of the 10 best in } D_j}$$

$\text{DCG} = \sum_{r \le 10} g(q_j, d_r) / \log_2(r+1)$

×100, mean over queries, then tasks

Pairwise

Bradley–Terry over all s_A, Elo scale

1 Qwen3-Embedding-8B
2 Qwen3-Embedding-4B
3 Linq-Embed-Mistral
⋮ 28 models · $\rho = 0.95$ with MTEB